

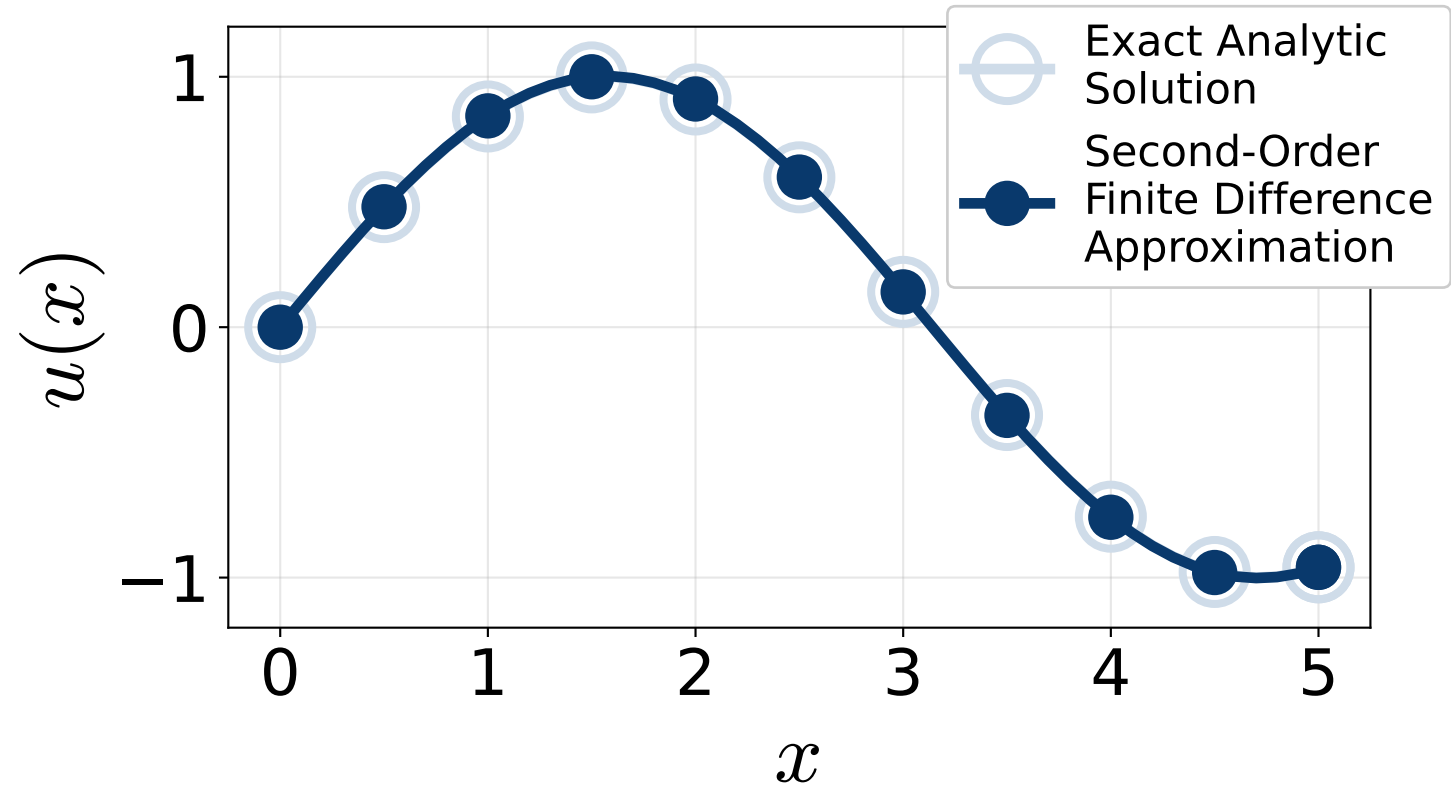


Figure 3.1: Accuracy of Second-Order Finite Difference Approximation of the test function $u = \sin(x)$ with $N = 50$.